

ExoSwarm

AI Mission Control for Agent-Orchestrated Exoplanet Investigation

Core concept

ExoSwarm is a multi-agent scientific investigation system that analyzes real astronomical observations, searches for exoplanet transit candidates, actively tries to falsify its own hypotheses, and produces a reproducible, evidence-based scientific disposition.

The key distinction remains:

ExoSwarm is not an LLM that detects planets.

Established astronomical algorithms perform the numerical work. The AI is the scientific decision-making and experiment-orchestration layer: it decides which observations or summarized evidence to inspect, which analysis should happen next, which competing astrophysical explanation is currently most plausible, what evidence would best distinguish between those explanations, and when the implemented evidence is sufficient to stop the investigation.

The stronger one-line pitch is:

ExoSwarm is an AI-orchestrated TESS investigation system that uses deterministic astronomy tools to test and falsify competing explanations for transit-like signals, then locks its measurements before comparing them with NASA ground truth.

For the shorter spoken version:

ExoSwarm investigates real NASA observations and tries to prove its own exoplanet hypotheses wrong.

This is stronger and more scientifically defensible than either "AI exoplanet detector" or "AI autonomously discovers confirmed planets."

The novelty claim should also be precise:

ExoSwarm does not invent new transit-detection mathematics. It turns established scientific computation into a visible, adaptive, adversarial investigation loop in which AI agents decide which scientific experiment should happen next.

The architecture deliberately combines a deterministic workflow with bounded agentic control:

ExoSwarm is a workflow for mandatory vetting and an agentic system for experiment selection. Baseline diagnostics are structural and code-enforced. Everything beyond that baseline is selected from the current evidence state by the agent layer.

That boundary is intentional. Fully unconstrained agents are neither necessary nor desirable for the scientific computation itself.

Why it fits the hackathon extremely well

The track asks for software applied to scientific computing, mathematical modeling, or biological/neuroscience analysis with meaningful use of AI and agents.

ExoSwarm directly satisfies the scientific-computing requirement because it processes astronomical time-series observations, performs numerical signal analysis, estimates physical parameters, tests competing astrophysical explanations, and compares independently calculated results with external scientific ground truth.

Its architectural principle is:

Agents decide what scientific operation should happen next; deterministic scientific software performs the operation.

That gives the project a strong answer in every major judging category:

- **Technical execution:** NASA/TESS data, FITS files, target-pixel data, numerical pipelines, external catalogs, model inference, structured state, tracing, tool execution, reproducibility tests and multiple APIs.
- **Advanced inference:** evidence changes which scientific experiment happens next rather than triggering a fixed one-pass chain.
- **Agent legitimacy:** agents have different objectives and restricted tool surfaces; they are not merely differently named prompts.
- **Scientific credibility:** numerical claims come from reproducible Python/scientific tools rather than model intuition.
- **Originality:** transit search and vetting already exist, but the differentiator is interactive adversarial orchestration, hypothesis-directed experiment selection and an observable reasoning workflow.
- **Utility:** ExoSwarm demonstrates how intelligent orchestration can sit above mature scientific software instead of replacing it.
- **Video demonstration:** real stellar observations transform into a detected signal, competing explanations, adaptive experiments, an evidence ledger, a locked prediction and an independent NASA reveal.
- **Repository verifiability:** judges do not need to trust an AI-generated score or a polished video. The blind protocol, traces, evals and reproducible result artifacts can be inspected directly in the source repository.
- **External verification:** judges can compare ExoSwarm's locked numerical prediction against an external catalog value.

Source: MAST / TESS data products documentation - <https://archive.stsci.edu/missions-and-data/tess/data-products>

The project therefore combines the four things most strategically useful for this hackathon:

real scientific computation + visible agentic inference + dramatic visualization + externally verifiable and reproducible results

Because this is an online hackathon with a recorded video rather than a live pitch, the project should optimize for **provable behavior rather than stage performance**. The video shows the system clearly; the repository proves that the behavior shown is real.

What actually happens inside ExoSwarm

A user selects a target from the mission-control interface.

The deterministic backend maps that real target to an opaque investigation identifier such as:

TARGET-X17

The scientific agents operate on that opaque identifier. They do not receive a recognizable TIC/TOI name, famous target name or other unnecessary metadata that could reveal the known answer through model training.

The backend privately maintains:

TARGET-X17 -> real TESS/TIC/TOI identity

and uses that mapping only where deterministic data access requires it.

For the hackathon demonstration, the exact TESS observations needed for the showcased targets are cached locally. This removes unnecessary MAST/Gaia/network risk without making the computation fake: the source files remain genuine NASA/TESS products with recorded provenance.

Each investigation has a structured state containing:

- opaque target identifier
- backend-only source mapping
- available data products
- data-quality information
- current hypotheses
- candidate signals
- completed experiments
- numerical evidence
- unresolved alternatives
- provenance
- tool versions and parameters
- remaining inference/experiment budget
- final lock status

The analysis proceeds roughly as follows.

1. Observation inspection

The Observer Agent examines compact data-quality diagnostics.

It does not inspect thousands of raw flux points inside the LLM context.

It decides things such as:

- whether the selected observation is scientifically usable
- whether the target contains excessive gaps or variability
- whether another permitted data product should be inspected
- whether the investigation should continue or terminate early

The broader architecture can support multiple sectors, but the hackathon demonstration deliberately uses a **small cached target suite and a single well-understood sector per showcased target**, preferably a suitable 2-minute SPOC product. Multi-sector stitching is outside the core 48-hour scope.

2. Signal preparation

The Signal Agent receives quality statistics and a limited set of preprocessing options.

Instead of always applying the same detrending recipe, it can select between allowed strategies depending on the observed data characteristics.

Deterministic Python performs every transformation.

3. Transit search

The Transit Hunter requests numerical transit-search operations.

Box Least Squares and related calculations return structured candidate signals including:

- candidate period
- epoch
- duration
- transit depth
- signal-to-noise
- number of observed events
- search statistics
- uncertainties or explicit precision/tolerance information where available
- artifact references for the periodogram and folded light curve

4. Competing hypotheses are created

A transit-like periodic signal does not immediately become a planet.

The investigation explicitly tracks alternatives such as:

- H1 - planetary transit
- H2 - eclipsing binary
- H3 - background/neighbor contaminant
- H4 - stellar variability
- H5 - instrumental/systematic artifact
- H6 - period alias or harmonic

5. Mandatory baseline diagnostics are enforced

A viable transit candidate automatically requires a structural baseline of scientific checks.

At minimum:

- minimum signal-quality checks
- odd/even comparison
- secondary-eclipse test
- basic contamination screening

These checks are not optional agent suggestions. They are code-enforced requirements.

6. The Skeptic selects a discriminating experiment

Beyond the mandatory baseline, the Skeptic Agent asks:

Which currently plausible alternative explanation could overturn our planetary interpretation, and which available experiment would most efficiently distinguish between them?

It chooses from a bounded experiment registry. It cannot invent arbitrary scientific analyses or unrestricted code.

Its decision is structured and includes:

- strongest unresolved alternative
- hypothesis under test
- requested experiment
- parameters
- reason code
- expected discriminating result
- predicted possible outcomes
- expected information value
- stopping condition if appropriate

The expected information value is an internal decision-quality estimate, not a scientific confidence score.

7. The Critic reviews the proposed experiment

Before an adaptive experiment executes, a bounded Critic Agent reviews the proposal against the current Evidence Ledger.

The flow is:

Skeptic proposes -> Critic reviews -> APPROVE | REVISE | VETO -> execute

The Critic asks one narrow question:

Would this experiment actually discriminate between the remaining hypotheses given the evidence already collected, or is it redundant or uninformative?

It may approve the experiment, request one revision, or veto a redundant choice and suggest one alternative.

Only one revision round is permitted before the runtime resolves the action. This creates a bounded evaluator-optimizer loop without allowing recursive agent debate.

8. Deterministic science executes the experiment

The validated request is passed to a deterministic Python scientific tool.

The LLM does not calculate the measurement itself.

9. Evidence updates the investigation

The tool result is appended to the Evidence Ledger.

The deterministic application layer converts tool outcomes into hypothesis-state updates using explicit, inspectable rules.

Where weighted hypothesis updates are used, they come from a declared evidence/likelihood table in the scientific specification rather than being invented by the model.

For example, an odd/even result can map to a declared change in the eclipsing-binary and planetary hypotheses.

These weights are treated as explicit model assumptions or heuristic evidence weights unless they have been statistically calibrated. They are **not displayed as calibrated planet probabilities merely because they are numeric.**

The core separation remains:

the agent chooses which experiment to perform; deterministic code owns what the experiment measured and how declared evidence rules update the investigation state.

10. Repeat

The system continues:

```
observe -> hypothesize -> choose experiment -> critique -> compute -> evaluate ->  
challenge -> update -> choose next experiment
```

until a defined stopping condition is reached.

11. Lock first, reveal second

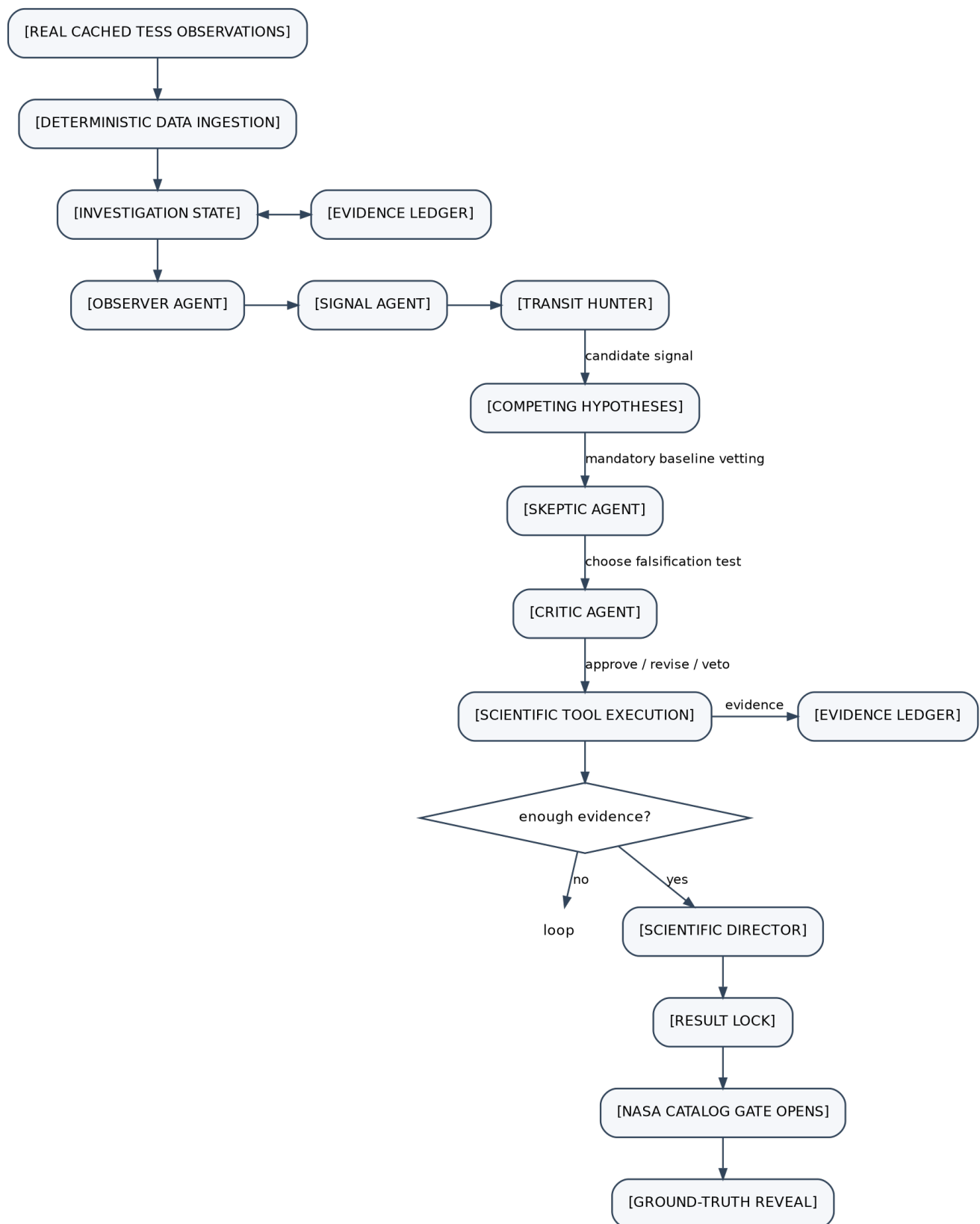
The final ExoSwarm result is serialized and locked.

Only after that event may the NASA ground-truth service become available.

The known answer is then revealed and quantitatively compared with the locked prediction.

This separation is enforced architecturally and in automated tests, not merely promised in the presentation.

A simplified architecture



A significant change is that the NASA catalog is not another freely available agent tool during the investigation.

It is a gated service.

Before `RESULT_LOCKED`, the scientific agents have no tool capable of returning the known planetary parameters for the target.

More importantly, they also do not receive a recognizable target identity unless a piece of identifying metadata is genuinely required for a scientific decision.

This strengthens the blindness protocol against model memorization.

The repository should make this separation mechanically verifiable:

- a test such as `test_catalog_unreachable_before_lock`
- a dependency/import boundary preventing the agent/science packages from directly importing the ground-truth catalog layer
- traces proving the lookup capability was unavailable before lock
- separate result and reveal artifacts

The blindness claim becomes a software invariant rather than a theatrical reveal.

Underneath the agents are real scientific functions

The underlying scientific toolkit should include:

- cached TESS light-curve loading
- cached Target Pixel File loading
- quality-flag inspection
- normalization
- outlier handling
- limited allowed detrending approaches
- Box Least Squares transit search
- phase folding
- candidate ephemeris calculation
- transit-depth measurement
- transit-duration estimation
- signal-to-noise measurement
- approximate R_p/R_{star} estimation
- odd-versus-even transit comparison
- secondary-eclipse search
- period-harmonic testing at $P/2$, P and $2P$
- stellar-neighbor/catalog context checks
- one genuine pixel-level / centroid contamination analysis
- uncertainty or tolerance estimation for reported measurements
- artifact and provenance recording

The original document correctly identified Lightcurve and the surrounding TESS ecosystem as the reason much of this is feasible during a 48-hour build.

TESS officially provides both light curves and Target Pixel Files, including aperture-mask and pixel information required for a credible contamination diagnostic.

Source: HEASARC TESS data products - <https://heasarc.gsfc.nasa.gov/docs/teess/data-products.html>

The pixel/centroid component should deliberately remain modest.

We are not trying to reproduce the complete probabilistic machinery of modern systems such as RAVEN in 48 hours. RAVEN already combines machine-learning classifiers, false-positive scenarios and positional information in a substantial TESS validation framework.

Source: RAVEN preprint - <https://arxiv.org/abs/2509.17645>

Our prototype needs one scientifically meaningful question:

Does the dimming event appear spatially consistent with the target star, or is there evidence that a nearby source may be responsible?

For the demonstrated target, the spatial diagnostic should use a cached Target Pixel File and produce a reproducible numerical and visual output, such as:

- out-of-transit image
- in-transit image
- difference image
- centroid shift
- offset with an uncertainty or significance estimate where scientifically justified

The result is recorded in the Evidence Ledger.

It must be described as a **limited contamination diagnostic**, not as a complete professional false-positive validation framework.

Tool contracts are part of the scientific architecture

Scientific tools should return more than unstructured numbers.

A result should look conceptually like:

```
odd_even_test
odd_depth_ppm: 1180
even_depth_ppm: 1340
difference_sigma: 0.7
interpretation_code: CONSISTENT
uncertainty: ...
limitations:
  - cannot exclude equal-depth eclipsing binaries
```

Every important tool should provide:

- numerical outputs
- uncertainty or precision information where applicable
- quality flags
- a bounded `interpretation_code`
- provenance
- limitations describing what the result cannot establish

Errors should also be scientifically useful.

Instead of:

`ValueError`

prefer a structured result such as:

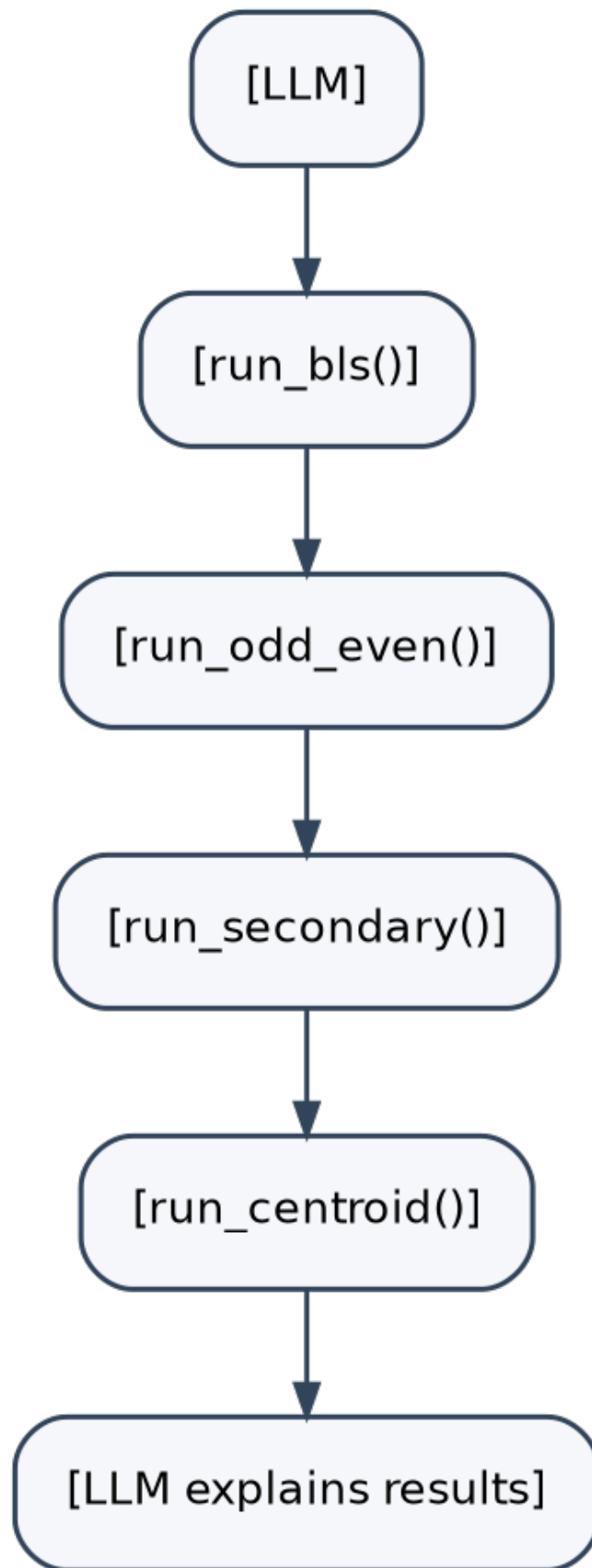
```
status: PRECONDITION_FAILED
reason: odd_even requires >=4 usable transits; found 3
suggested_alternatives:
  - secondary_eclipse
  - harmonic_test
```

This allows the agent to re-plan without inventing a workaround.

The AI / advanced inference pipeline

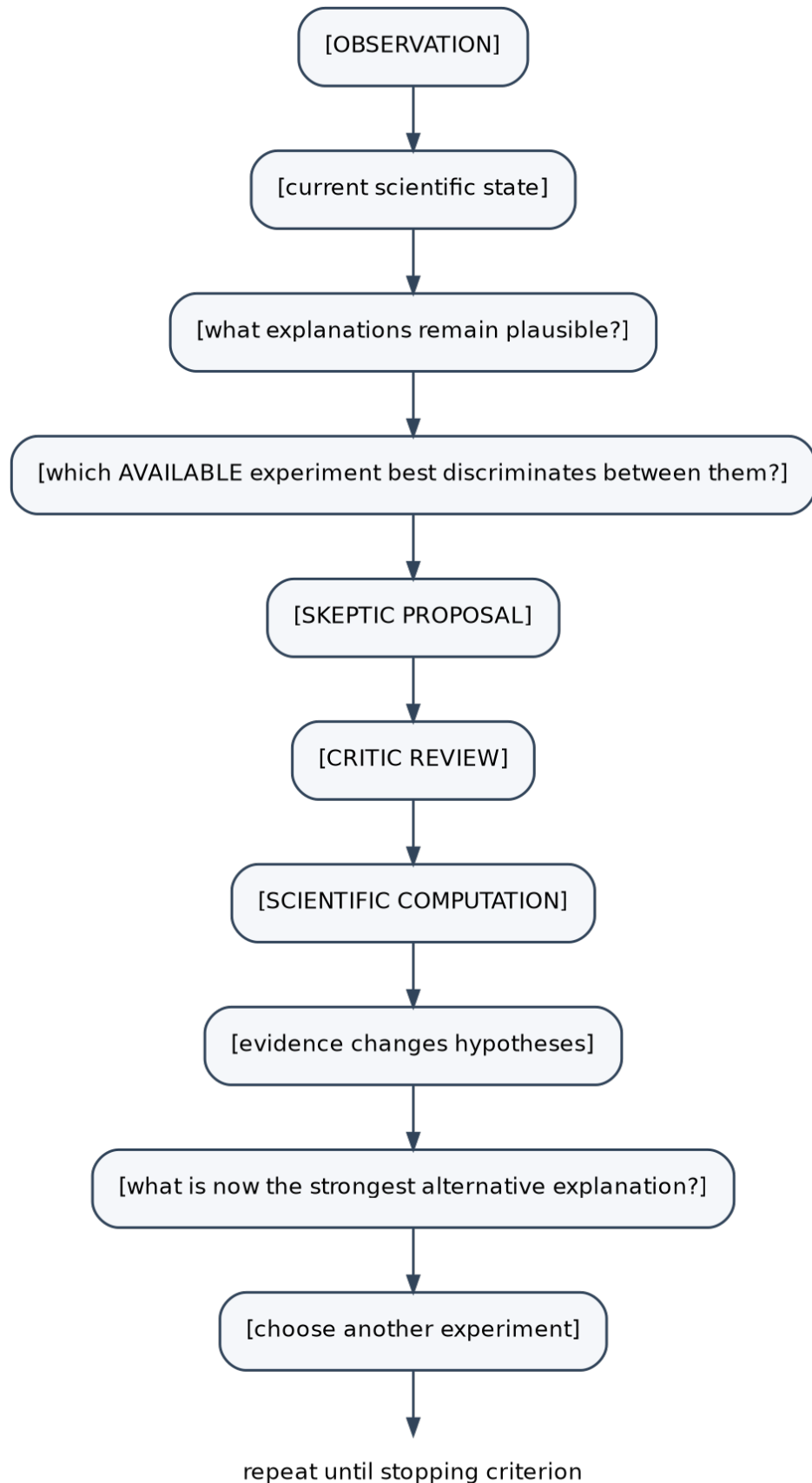
This remains the part that should be emphasized most strongly to judges.

A weak implementation is:



That is primarily a fixed Python pipeline with an LLM narrator.

ExoSwarm should instead behave like:



This preserves the original:

observe -> hypothesize -> test -> falsify -> update

idea, which is still the strongest conceptual component of the project.

Mandatory diagnostics versus adaptive diagnostics

We should not allow scientific safety checks to disappear merely because an LLM forgot about them.

Therefore ExoSwarm uses two layers.

Mandatory baseline vetting

A viable transit candidate automatically requires:

- minimum signal-quality checks
- odd/even comparison
- secondary-eclipse test
- basic contamination screening

These are structural requirements.

Adaptive experiments

Beyond them, the Skeptic decides what should happen next.

Case A - suspicious odd/even difference

Evidence:

```
odd_transit_depth != even_transit_depth
```

Skeptic:

Eclipsing-binary hypothesis strengthened. Test whether the detected period is a half-period alias.

Next experiment:

```
rerun candidate analysis at 2P
```

Case B - odd/even clean, but nearby star exists

Evidence:

```
neighbor inside/near TESS aperture
```

Skeptic:

Background-source hypothesis now dominates. Spatial localization is more informative than another light-curve test.

Next experiment:

```
run_centroid_localization()
```

Case C - low SNR and strong stellar variability

Evidence:

- periodogram candidate exists
- high local variability

- transit shape unstable

Skeptic:

Current signal may depend on preprocessing.

Next experiment:

repeat using another allowed detrending configuration.

Case D - significant secondary event

Evidence:

secondary-like feature near phase 0.5

Skeptic:

Eclipsing-binary hypothesis materially strengthened.

Next experiment:

compare primary/secondary depths and inspect the period harmonic.

These branches are the proof that the agents change the analysis.

Current best-practice agent architecture

The runtime should not be implemented as several unrestricted LLMs talking endlessly to one another.

ExoSwarm uses a **manager + bounded-specialist architecture**.

The architecture can be described using established agent patterns:

- **Manager / agents-as-tools:** the Scientific Director retains control while specialists perform bounded decisions.
- **Orchestrator-workers:** the Director delegates scientific decision subtasks according to the current investigation state.
- **Evaluator-optimizer:** the Skeptic proposes an adaptive experiment and the Critic reviews its scientific usefulness before execution.

The important point is not that ExoSwarm has many named agents.

The important point is that each bounded agent has a different optimization target, context and permitted action surface.

The Scientific Director is the manager

The Director owns the current investigation.

It can invoke specialist agents for bounded decisions:

- Observer Agent
- Signal Agent
- Transit Hunter

- Skeptic Agent
- Critic Agent

The specialists do not own the complete conversation and do not need every previous token.

They receive only the evidence necessary for their task and return a structured result.

The Director combines those decisions with deterministic application logic.

Additional agents and handoff-heavy architectures are not part of the core hackathon scope.

Use structured agent outputs

Do not let important decisions be arbitrary prose.

For example:

```
SkepticDecision

hypothesis_under_test:
  background_contamination

requested_experiment:
  centroid_localization

parameters:
  candidate_period: <ledger reference>

reason_code:
  NEARBY_SOURCE_IN_APERTURE

expected_discriminating_result:
  determine whether transit-associated centroid motion
  is consistent with the target position

predicted_outcomes:
  TARGET_CONSISTENT: ...
  OFFSET_DETECTED: ...

expected_information_value:
  ...

stop_if:
  spatial evidence resolves the dominant remaining alternative

priority:
  HIGH
```

The model may generate a short human-readable explanation for the UI, but the application consumes the structured object.

The model's predicted outcomes and expected information value are evaluated as **agent-decision quality signals**, not as measured astrophysical probabilities.

After the tool executes, the deterministic system can compare the expected discriminating value against the actual change in the declared hypothesis state.

This lets ExoSwarm evaluate not merely whether an agent chose *an* experiment, but whether it tended to choose informative experiments.

The outer loop belongs to code

The runtime should have an explicit loop approximately like:

```
initialize investigation

while investigation.status == ACTIVE:
    assemble compact evidence packet
    ask appropriate agent for next structured decision
    validate structured output
    validate requested experiment

    if adaptive experiment:
        ask Critic to APPROVE / REVISE / VETO
        allow at most one revision round

    execute deterministic tool
    validate scientific output
    append output to evidence ledger
    deterministically update hypothesis state

    record:
        decision
        critic verdict
        expected information value
        resulting evidence change

    check:
        mandatory tests completed?
        discriminating experiment still available?
        sufficient evidence?
        experiment budget exhausted?
        max turns reached?
        repeated low-value action?

lock result
```

The LLM supplies scientific choices.

Code controls:

- permissions
- maximum turns
- valid tools
- required diagnostics
- budgets
- state transitions
- catalog isolation
- belief/evidence update rules
- failure recovery
- lock semantics

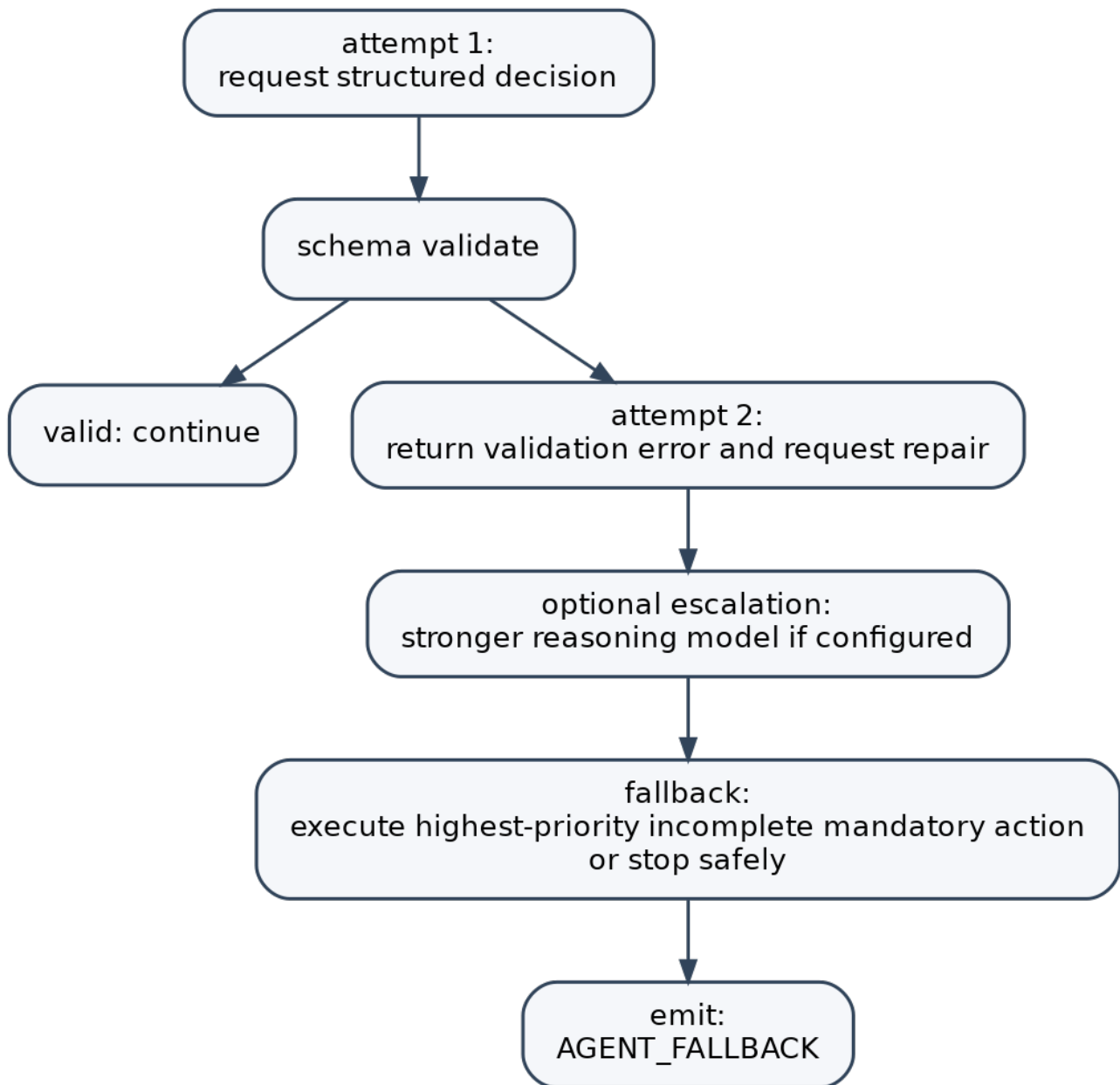
This is both more scientifically defensible and easier to audit.

Structured-output reliability

The agent layer must have an explicit failure policy.

A model request should never silently fall through into arbitrary behavior.

Conceptually:



Every fallback is recorded in the trace.

The project should be able to report:

- schema-valid output rate
- repair rate
- escalation rate if model routing is enabled
- fallback rate

A judge reading the repository should be able to see that failure handling is declared rather than hidden.

Harness engineering for ExoSwarm

The scientific Python layer should remain explicit application code rather than giving the model unrestricted shell access to invent analyses.

Use the agent runtime for:

- agent execution
- specialist invocation
- structured outputs
- tool schemas
- traces
- guardrails
- model usage tracking
- lifecycle events

Use explicit Python functions for:

- astronomical computation
- data retrieval/loading
- plotting
- state mutation
- evidence calculation
- hypothesis-state updates
- target locking
- NASA catalog gating

Tool guardrails

A requested scientific experiment should be validated before execution.

Examples:

- reject unknown tools
- reject nonsensical period ranges
- reject catalog access before lock
- reject modifications to raw source data
- enforce parameter ranges
- prevent duplicate expensive experiments unless justified
- cap experiment count
- enforce tool preconditions
- prevent a tool result from claiming more than its contract allows

A numeric-provenance guardrail should also protect UI text.

Agent-generated scientific prose can be checked so that numerical values appearing in the interface must correspond to values already present in the Evidence Ledger or another deterministic artifact.

Conceptually:

GuardrailViolation:
agent output contained "4.2 hours",
but no corresponding measurement exists in the ledger.

This changes:

"The LLM is instructed not to invent measurements."

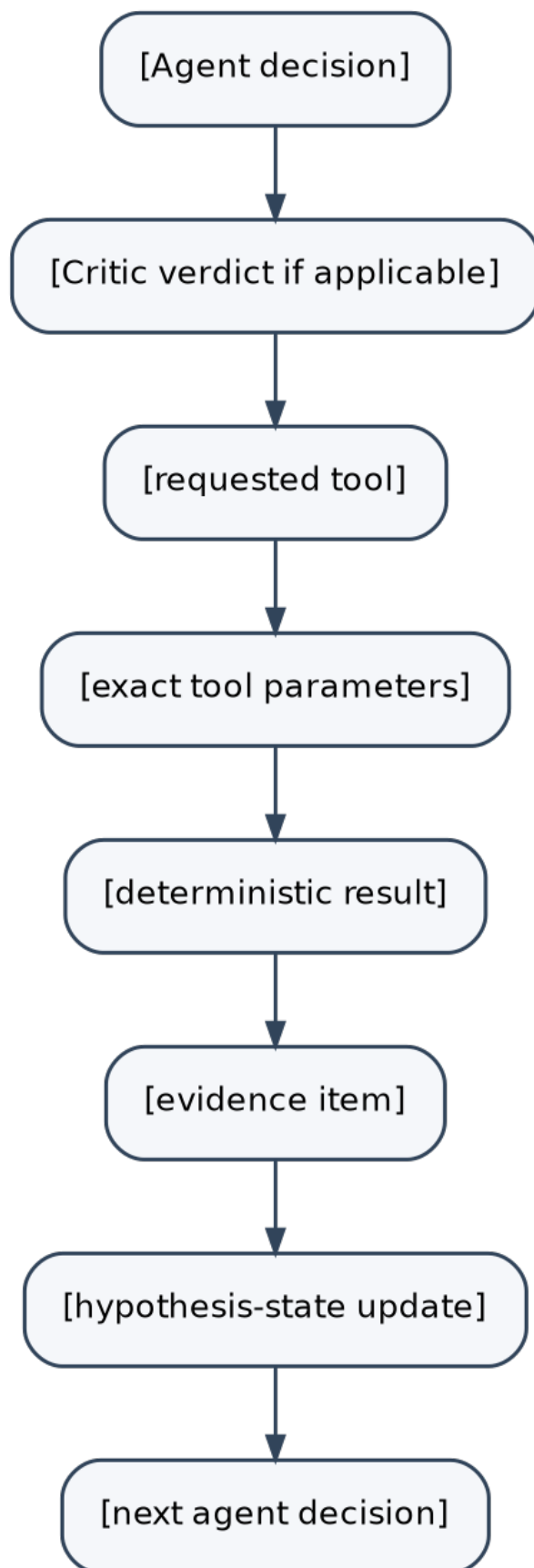
into:

"The application mechanically rejects unsupported numerical claims."

Trace everything

Every investigation should have a trace ID.

For debugging and judging, we should be able to reconstruct:



The trace should additionally record:

- target opaque identifier
- token usage
- structured-output failures

- fallback events
- expected information value
- realized evidence-state change
- result-lock event
- exact point at which catalog access becomes available

That trace is both a development tool and evidence that the project is genuinely agentic.

Context engineering

This is one of the most important implementation choices.

Do not send tens of thousands of raw light-curve samples into the LLM.

Python processes them and returns compact scientific results.

The model should **not** receive:

- 20,000 raw flux measurements
- complete FITS files
- entire TESS headers
- every previous plotting object
- the complete transcript of every other agent
- raw Target Pixel File arrays
- known catalog parameters before result lock
- recognizable target identity unless scientifically necessary

The application should maintain locally a typed `InvestigationState`, preferably using Pydantic/dataclasses, containing everything required by the runtime.

Example:

```
InvestigationState
|- opaque_target_id
|- backend_target_mapping
|- data_artifacts
|- preprocessing_runs
|- candidates
|- hypotheses
|- evidence
|- completed_tests
|- available_tests
|- unresolved_questions
|- provenance
|- inference_budget
`- lock_state
```

Agents receive compact evidence packets

For example:

```
TARGET
TARGET-X17

CURRENT CANDIDATE
period = 6.26814 +/- ...
epoch = ...
depth = 1320 ppm +/- ...
duration = 2.41 h +/- ...
SNR = 12.8
n_transits = 7

COMPLETED TESTS
signal_quality: PASS
odd_even: PASS
secondary_eclipse: PASS

NEW EVIDENCE
neighbor detected inside aperture context
delta_mag = 2.1

OPEN HYPOTHESES
planetary transit
background eclipsing source

AVAILABLE EXPERIMENTS
centroid_localization
alternate_aperture_test
stop
```

This is enough for a sophisticated scientific decision without drowning the model in irrelevant data.

The trace should record context size per decision.

A useful evaluation statistic is:

- median tokens supplied per agent decision
- maximum tokens supplied
- confirmation that raw light curves never entered model context

For this bounded 8-15-turn style of scientific investigation, elaborate long-horizon compaction and autonomous note-taking are unnecessary core features.

Progressive disclosure

If the Skeptic needs more information, it should request it.

For example:

```
get_evidence("odd_even_test_03")
get_plot_summary("phase_folded_candidate_01")
get_neighbor_summary()
```

Do not eagerly load everything.

Context isolation between agents

The Transit Hunter does not need the complete Skeptic prompt.

The Skeptic does not need every intermediate detrending array.

The Critic needs the current proposal and relevant ledger state, not every previous message.

The Observer does not need NASA catalog records.

Each specialist gets:

1. its objective
2. its permitted tools
3. compact state relevant to that objective
4. a small set of canonical examples
5. a strict structured-output schema

This prevents the multi-agent architecture from degenerating into several copies of the same model with the same giant context.

Prompt design

Avoid enormous persona prompts.

Each agent prompt should contain approximately:

```
ROLE  
SCIENTIFIC OBJECTIVE  
WHAT YOU MAY ASSUME  
WHAT YOU MUST NOT CLAIM  
AVAILABLE TOOLS  
DECISION RULES / HEURISTICS  
STOP / ESCALATION CONDITIONS  
STRUCTURED OUTPUT FORMAT  
2-4 CANONICAL EXAMPLES
```

For example, the Skeptic prompt should not simply say:

```
"You are a skeptical astronomer."
```

It should say, in substance:

```
Your objective is to identify the strongest plausible non-planetary explanation  
that remains compatible with the evidence and select the available experiment  
expected to best discriminate between it and the planetary hypothesis. Do not  
invent measurements. Do not assign scientific probabilities unless they are  
supplied by a defined deterministic/statistical method. Prefer experiments that  
can change the disposition. Return exactly one requested experiment or STOP.
```

The Critic has a different objective:

Evaluate whether the proposed experiment is genuinely discriminating given the evidence already collected. Reject redundant or uninformative actions. Return APPROVE, REVISE or VETO and at most one alternative.

These produce genuinely different optimization targets.

Model strategy

ExoSwarm will use the Featherless AI API as its model provider.

The architecture remains model-agnostic: agents interact with a common inference interface, while the rest of the scientific pipeline, state, tools, schemas, Evidence Ledger, evaluation harness and UI do not depend on a particular model provider.

For the **core hackathon build**, model routing is deliberately lower priority than scientific reliability.

The system should first work reliably with one primary model.

DeepSeek-V4-Flash-0731

This is the intended default model for the majority of agentic workload, including:

- primary orchestration
- Observer Agent decisions
- Signal Agent decisions
- Transit Hunter tool selection
- ordinary Skeptic decisions
- Critic review
- interpreting structured deterministic tool outputs
- concise mission-control explanations
- deciding whether additional evidence is required
- routine structured-output generation

Most ExoSwarm decisions occur over a compact and highly structured scientific state and involve choosing among clearly defined tools.

The core goal is therefore not to use the strongest model available for every call.

It is to use a model that reliably performs the bounded scientific decision.

Optional model escalation - stretch goal

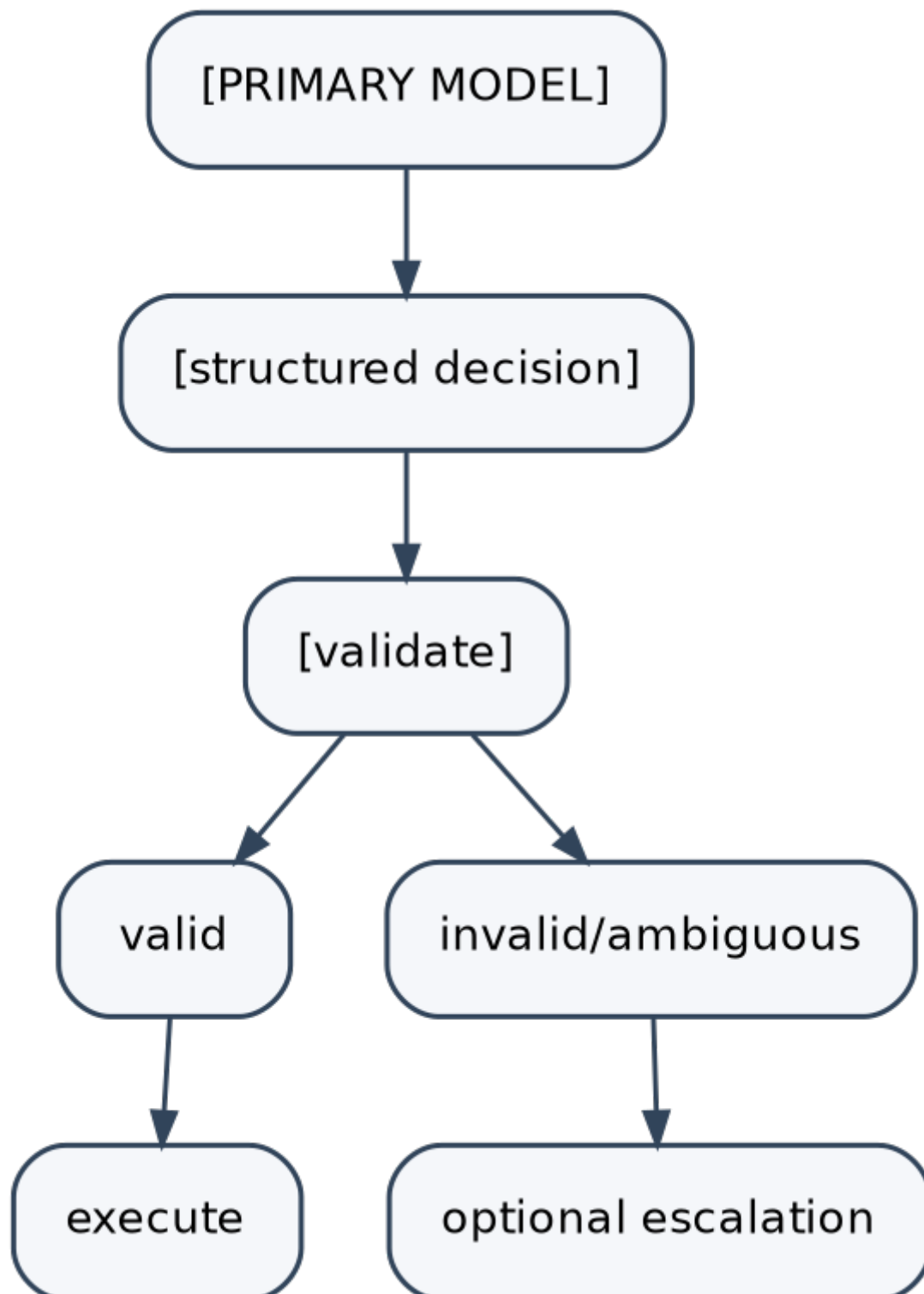
GLM-5.2 can be added selectively as a higher-reasoning model only after the complete core investigation loop works reliably.

Possible uses include:

- difficult Skeptic decisions
- several plausible competing hypotheses
- ambiguous false-positive cases
- contradictory evidence

- difficult final synthesis
- repair after repeated invalid primary-model decisions

Conceptually:



Model routing is therefore **P2**, not a condition for the core demonstration.

If it is implemented, models should be routed based on eval results rather than assumptions.

Measure:

- valid tool selection
- correct hypothesis handling
- unnecessary tool calls
- repeated experiments
- scientifically invalid claims
- structured-output failures
- trajectory quality
- latency
- token usage
- cost

The objective remains:

Use the cheapest model that reliably makes the required scientific decision, and spend additional inference only where the investigation genuinely benefits from it.

Models never become the scientific authority.

Neither the primary model nor an optional escalation model should generate scientific measurements from intuition.

They receive deterministic evidence such as:

```
CandidateSignal
period_days: 6.26812 +/- ...
epoch_btjd: ...
duration_hours: 2.41 +/- ...
depth_ppm: 1320 +/- ...
snr: 12.8
n_transits: 7
```

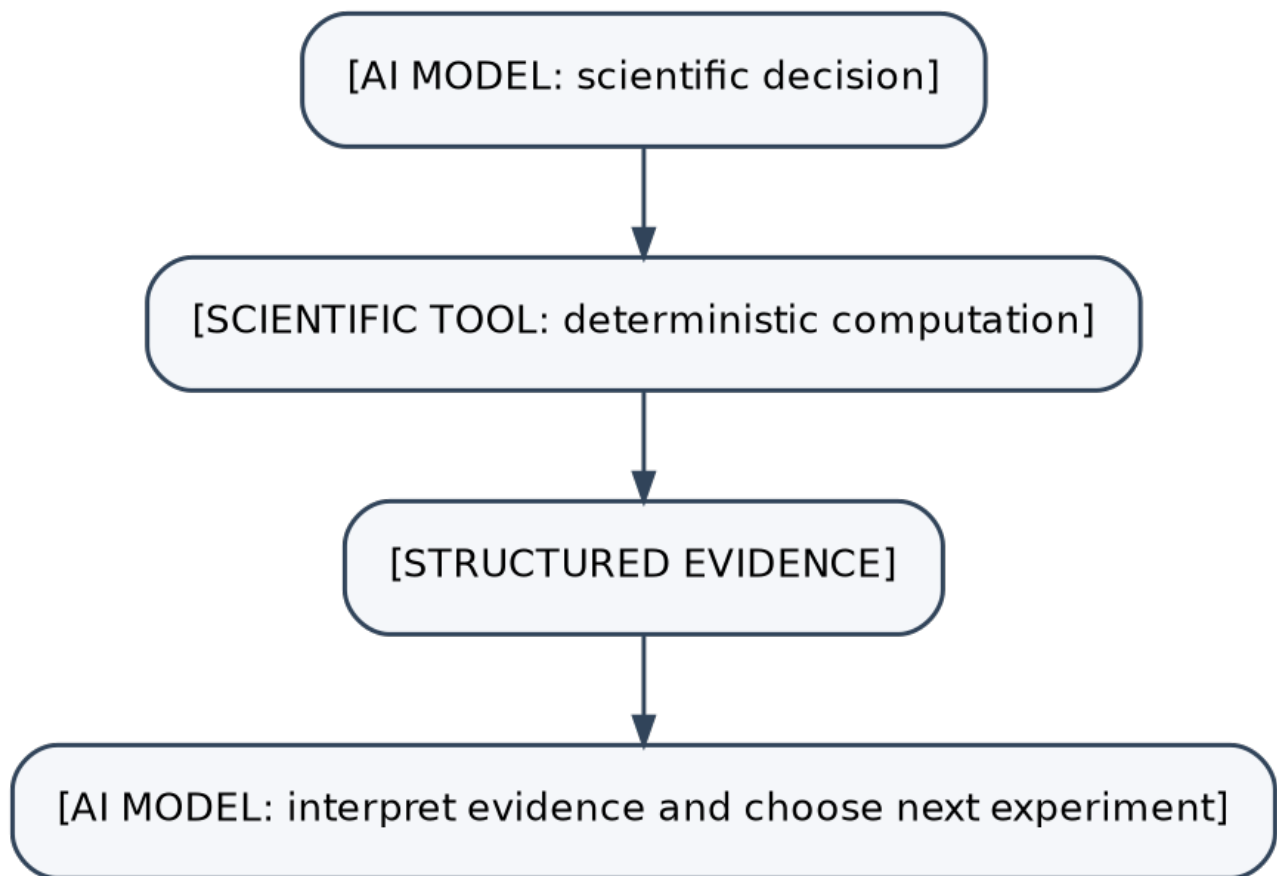
and reason about:

What should we test next?

They do not decide:

What numerical value should the measurement have?

The separation remains:



Inference budget

Because raw astronomical observations remain outside the LLM context and scientific functions return compact structured evidence, model calls should remain relatively small.

The inference budget should be concentrated on:

- experiment selection
- hypothesis comparison
- falsification reasoning
- difficult branch decisions
- Critic review
- final synthesis

Not on:

- processing raw light curves
- calculating numerical statistics
- rendering plots
- manipulating FITS files
- performing BLS
- measuring transit depths
- computing odd/even differences
- performing centroid calculations

Those remain deterministic Python operations.

Agent evaluation harness

The project should include a small internal eval suite.

This is especially important in an online hackathon because judges cannot interactively challenge the system on stage.

A beautiful video of one target does not prove the policy works.

Create roughly 6-10 curated evaluation cases, including:

- several clean confirmed planets
- one or more eclipsing binaries
- one contaminated candidate
- one noisy/variable target
- one deliberately difficult or inconclusive case

We do not need to showcase all of them.

They exist to prevent optimizing the agent policy for one demo star.

Grade outcomes, not a single prescribed path

Useful deterministic graders include:

- Did catalog access occur before `RESULT_LOCKED`? PASS/FAIL
- Did recognizable target identity enter the agent context before reveal? PASS/FAIL
- Was a measurable candidate recovered? PASS/FAIL
- Is period error below the defined target tolerance? PASS/FAIL
- Were mandatory tests performed? PASS/FAIL
- Did scientifically different targets produce different investigation branches? PASS/FAIL
- Did a known false-positive target avoid a strong planetary disposition? PASS/FAIL
- Did an inconclusive target remain inconclusive when appropriate? PASS/FAIL
- Did any agent invent a numerical measurement without an evidence/tool reference? PASS/FAIL
- Was the maximum agent-turn limit respected? PASS/FAIL
- Were tool-call schemas valid? PASS/FAIL

Do **not** require a specific adaptive tool to occur at an arbitrary numbered step merely because that is what happened once.

The agent should be graded primarily on whether the final scientific state and required constraints are correct.

Test both directions

The evaluation suite should contain cases where a behavior should occur and cases where it should not.

For example:

- a target where harmonic testing is appropriate
- a target where harmonic testing would be unnecessary
- targets whose planetary interpretation survives implemented vetting
- targets that should be rejected/downgraded
- at least one target that should remain inconclusive

This prevents one-sided optimization.

Measure consistency

For stochastic agent decisions, run important eval cases multiple times.

Report a consistency metric such as `pass^3`: whether the target succeeds correctly across three independent runs.

For a scientific system, consistency matters more than eventually obtaining one successful trajectory.

Agent-vs-fixed-policy ablation

This is a major competitive advantage.

Create a simple fixed baseline such as:

```
BLS -> odd/even -> secondary -> centroid -> final
```

Then run the same curated scientific states through the adaptive ExoSwarm selector.

Compare:

- correct next experiment where a best action is definable
- correct final disposition
- unnecessary experiments
- repeated tool calls
- invalid tool requests
- mandatory-test completion
- mean experiments run
- realized information value per adaptive experiment
- turns
- latency
- cost
- consistency across repeated runs

The purpose is **not** to prove that LLM agents outperform professional astronomy pipelines.

The purpose is to establish that the agent layer meaningfully controls the investigation trajectory and can seek evidence more selectively than a fixed demonstration checklist.

If the adaptive and fixed systems perform essentially identically, the project should report that honestly rather than manufacturing a benefit.

Repository-level reproducibility

Because this is an online submission, the repository should contain direct evidence for the major claims shown in the video.

For each showcased or reproducible investigation, keep artifacts conceptually like:

```
runs/  
  TARGET-X17/  
    result.json  
    result.json.sha256  
    trace.jsonl  
    reveal.json  
  
evals/  
  report.md
```

`result.json` is written and hashed before the reveal.

`trace.jsonl` contains the scientific trajectory.

`reveal.json` is produced only after lock.

A command such as:

```
make reproduce
```

should be able to rerun the deterministic investigation from cached source data without depending on astronomy-data network access and re-check the produced artifacts.

This turns reproducibility into part of the product rather than merely a claim.

The scientific computation must remain the authority

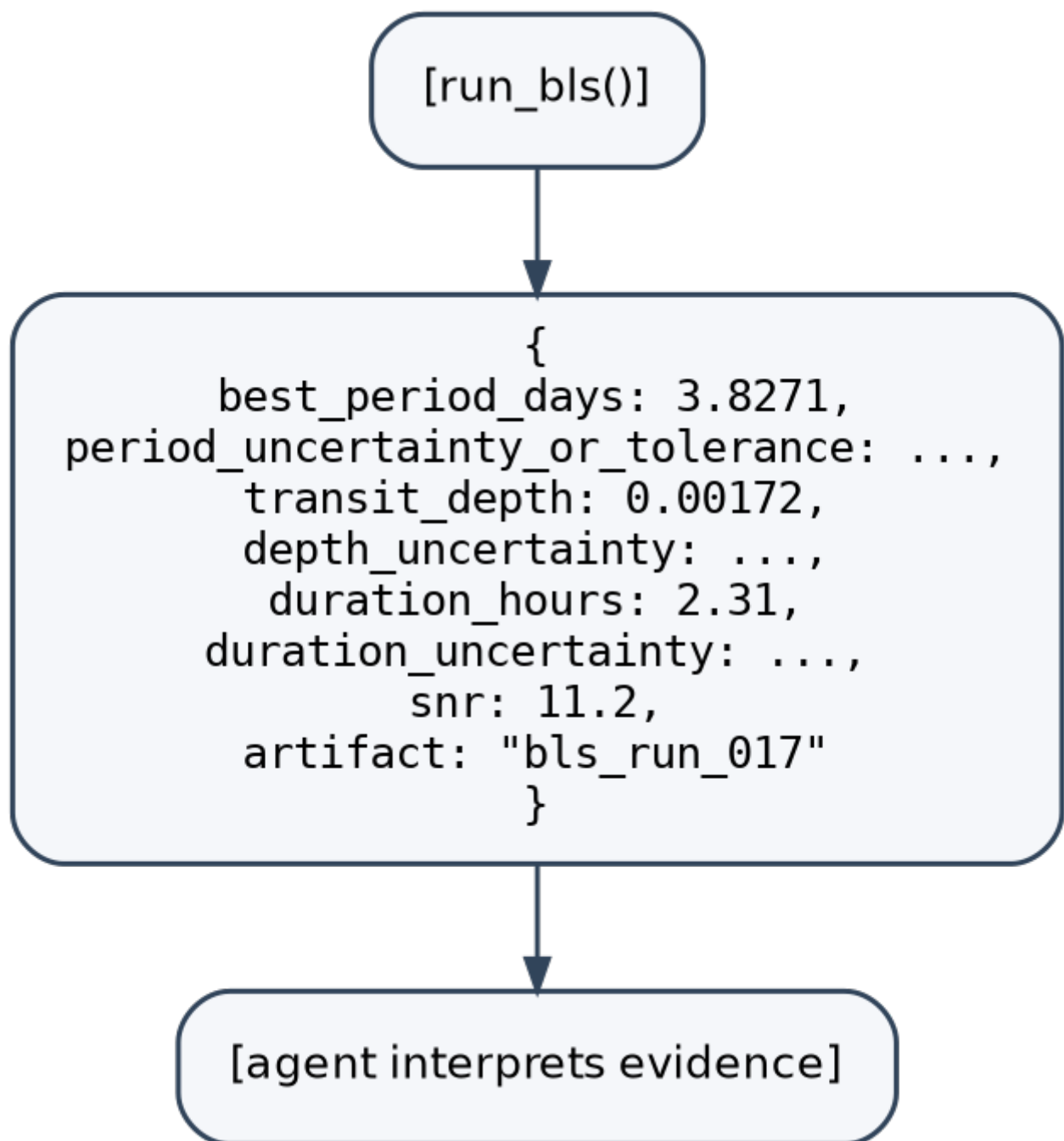
This remains non-negotiable.

The LLM should never say:

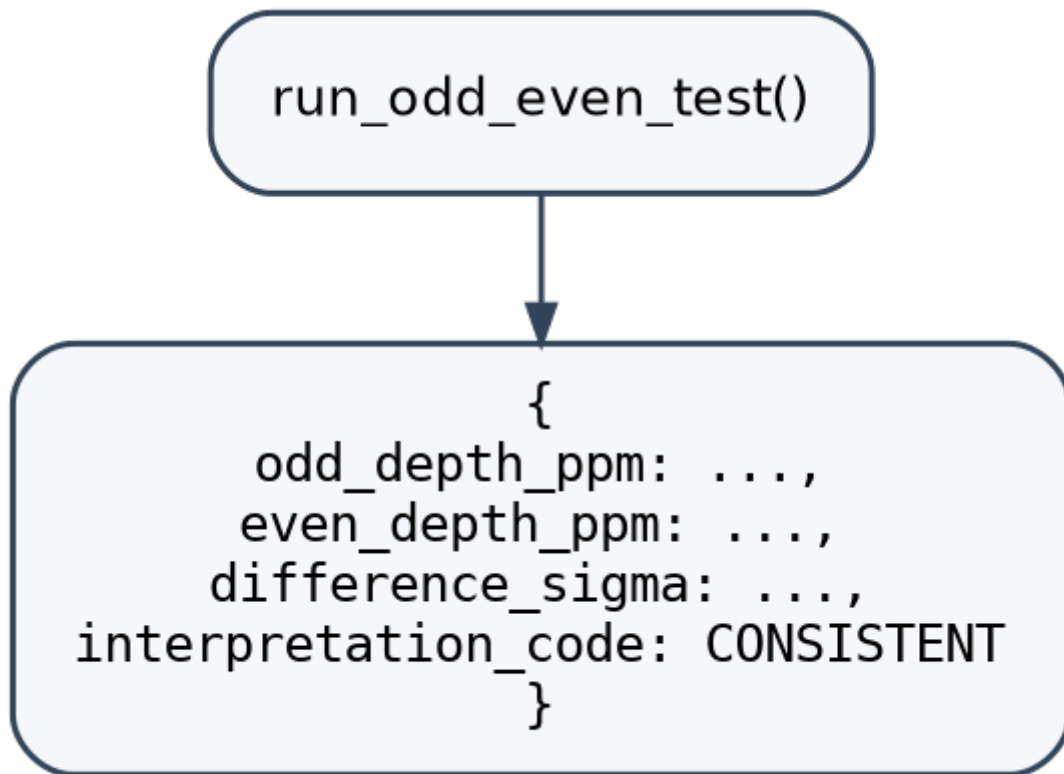
```
"I estimate the orbital period is 3.82 days"
```

because it visually inspected the curve.

Instead:



Likewise:



The model interprets measurements.

It does not manufacture measurements.

This makes ExoSwarm cheaper, faster, more reproducible and easier to defend.

The Evidence Ledger

Add an append-only Evidence Ledger to every investigation.

Every scientific result should record something close to:

```
EvidenceItem
id
experiment_type
tool_name
tool_version
input_artifact_ids
input_hash
parameters
numerical_results
uncertainties
interpretation_code
quality_flags
limitations
output_artifact_ids
timestamp
agent_request_id
critic_decision_id
```

Hypothesis disposition should be derived from these objects plus declared deterministic update rules.

For the UI, convert them into concise statements:

```
+ periodicity detected
  P = 6.26814 d +/- ...

+ odd/even transit depths consistent
  difference = 0.4 sigma

+ no significant secondary event detected

! nearby source inside contamination radius

+ centroid motion consistent with target
  offset = ... +/- ...

? low-SNR secondary remains unconstrained
```

This Evidence Ledger becomes the spine of the entire product.

It powers:

- agent context
- reproducibility
- audit trails
- UI storytelling
- final disposition
- eval grading
- video credibility
- numeric provenance guardrails

Do not use fake AI confidence

Remove:

Confidence: 94%

and remove arbitrary labels such as:

Confidence: HIGH

if the score is merely model-generated.

Modern TESS validation systems already perform explicit probabilistic modeling. ExoSwarm should not invent a worse probability merely because percentages look impressive.

Instead display:

CURRENT DISPOSITION

PLANETARY TRANSIT HYPOTHESIS SURVIVES IMPLEMENTED VETTING

Supporting evidence

- + strong periodicity
- + repeated transit morphology
- + odd/even consistency
- + no significant secondary
- + spatial test consistent with target

Remaining uncertainty

! photometric vetting alone does not constitute confirmation

Possible disposition states:

- NO CREDIBLE PERIODIC SIGNAL
- TRANSIT-LIKE SIGNAL
- PLANETARY INTERPRETATION WEAK
- PLANETARY INTERPRETATION PLAUSIBLE
- PLANETARY INTERPRETATION SURVIVES IMPLEMENTED VETTING
- INCONCLUSIVE - ADDITIONAL DATA REQUIRED

If the deterministic hypothesis-update layer uses numeric evidence weights, those values remain internal, declared and auditable.

They must not be presented as calibrated scientific probability unless the statistical interpretation is genuinely justified.

Add uncertainties to final measurements

BLS is excellent for locating a candidate but a long decimal string can imply more physical precision than the analysis supports.

The final locked artifact should therefore communicate uncertainty.

Use BLS to locate period, epoch and duration.

If time permits, fit a lightweight trapezoid or simple transit model around the candidate.

Report:

- period with uncertainty or explicit tolerance
- epoch with uncertainty or explicit tolerance
- depth with uncertainty
- duration with uncertainty
- approximate radius ratio, clearly labeled approximate unless the fitting method justifies more

The project should prefer scientifically honest precision over visually impressive decimal places.

Why hasn't someone already made exactly this?

Most individual pieces already exist.

The prior-art section must be explicit.

Existing scientific pipelines already do much of the underlying science

SHERLOCK is an end-to-end pipeline that can retrieve observations, search for transit signals, vet promising signals, perform statistical validation and refine planetary parameters.

Source: SHERLOCK paper - <https://arxiv.org/abs/2407.14602>

RAVEN performs sophisticated automated TESS vetting and validation against multiple false-positive scenarios.

Source: RAVEN preprint - <https://arxiv.org/abs/2509.17645>

Therefore:

We should never claim that ExoSwarm invented automated exoplanet transit detection or automated vetting.

Agentic exoplanet science also exists

ASTER, introduced in 2026, is explicitly an Agentic Science Toolkit for Exoplanet Research and uses LLM orchestration over domain scientific tools for atmospheric-characterization workflows.

Source: ASTER - <https://arxiv.org/abs/2603.26953>

Stargazer evaluates AI agents on iterative, physics-grounded exoplanet inference using radial-velocity data and repeated scientific model fitting.

Its results are also a useful warning: an agent can obtain numerically good results while still failing to recover the correct physical explanation.

Source: Stargazer - <https://arxiv.org/abs/2604.15664>

That warning reinforces ExoSwarm's design:

Deterministic numerical success is evidence, not permission for the LLM to make an unconstrained physical claim.

What remains differentiated

What does not appear to exist as a mature mainstream product is this specific combination:

TESS transit investigation + adaptive LLM experiment selection + explicit adversarial falsification + deterministic Evidence Ledger + bounded Critic review + visible scientific decision-making + provably blinded catalog reveal + mission-control UX

That is the originality claim.

Not the algorithms.

Not the fact that AI agents exist.

The workflow and interaction model.

Why might this not already be a major scientific product?

1. Deterministic pipelines are better for bulk processing

Astronomers do not need an LLM to calculate a periodogram for millions of stars.

Traditional software is cheaper, more reproducible and vastly easier to run at scale.

So ExoSwarm should not compete with that.

Its useful layer is:

adaptive investigation of interesting cases after deterministic pipelines expose useful evidence.

2. Agent orchestration has only recently matured

Current agent frameworks provide structured tool use, persistent application state, guardrails, tracing and controlled loops that make this class of application easier to implement reliably than several years ago.

3. Scientific reproducibility matters

Researchers prefer workflows whose measurements and assumptions can be reconstructed.

That is why ExoSwarm has:

- deterministic scientific tools
- declared hypothesis-update rules
- an append-only Evidence Ledger
- reproducible result artifacts

4. Additional observations may still be required

A transit-like event may be:

- an eclipsing binary
- contamination
- stellar variability
- instrumental noise
- another astrophysical configuration

No AI can infer spectroscopy or radial-velocity measurements that were never collected.

The project therefore stops at **candidate investigation and implemented vetting**, not autonomous confirmation.

5. Agents can fail recursively

More model inference does not automatically produce better physical reasoning, and agents can fall into repetitive failure loops.

That is why ExoSwarm uses:

- a bounded experiment registry
 - structured state
 - maximum experiment count
 - mandatory diagnostics
 - Critic review
 - explicit stopping criteria
 - deterministic graders
 - an eval suite
 - structured-output fallback behavior
-

Can it actually produce real results?

Yes.

This remains one of the project's strongest reasons to exist.

The blinded-test concept should be retained and strengthened.

A carefully selected confirmed TESS exoplanet is treated as a hold-out experiment.

The deterministic backend receives the target mapping required to load the cached TESS data.

The **agents** receive:

- an opaque identifier such as `TARGET-X17`
- compact observation/evidence summaries
- ordinary scientifically necessary context that does not reveal the known planetary solution

They do not receive:

- the real target name if avoidable
- recognizable identifiers if avoidable
- known planetary parameters
- NASA confirmation status
- ground-truth catalog results

The system independently estimates:

- orbital period
- transit epoch
- transit duration
- transit depth
- signal-to-noise
- approximate radius ratio
- odd/even consistency

- secondary-event evidence
- contamination evidence
- evidence for or against the planetary interpretation

Example final artifact:

```
EXOSWARM LOCKED RESULT

Target:
TARGET-X17

Candidate period:
6.26812 d +/- ...

Transit epoch:
... +/- ...

Transit duration:
... +/- ...

Transit depth:
... +/- ...

Odd/even:
consistent

Secondary event:
not significant

Spatial contamination:
no significant displacement detected

Disposition:
planetary interpretation survives implemented vetting

SHA256(result.json):
8f45...
```

Only then does the ground-truth service unlock.

```
TARGET REVEAL
Real target identity:
...

NASA EXOPLANET ARCHIVE

Known period:
6.26794 d

EXOSWARM:
6.26812 d

Absolute error:
0.00018 d

Relative error:
0.0029%
```

The NASA Exoplanet Archive provides programmatic resources that make this comparison straightforward to automate.

Source: NASA Exoplanet Archive API resources - [https://exoplanetarchive.ipac.caltech.edu/docs/API_resources.html](https://exoplanetarchive.ipac.caltech.edu/docs/API_resources.html)

The important sentence is:

The catalog is an evaluator, not an input to the investigation.

Make the blindness technically provable

Do not merely hide the NASA result in the UI.

Use separate capabilities.

Before lock:

AGENT-VISIBLE IDENTITY
TARGET-X17

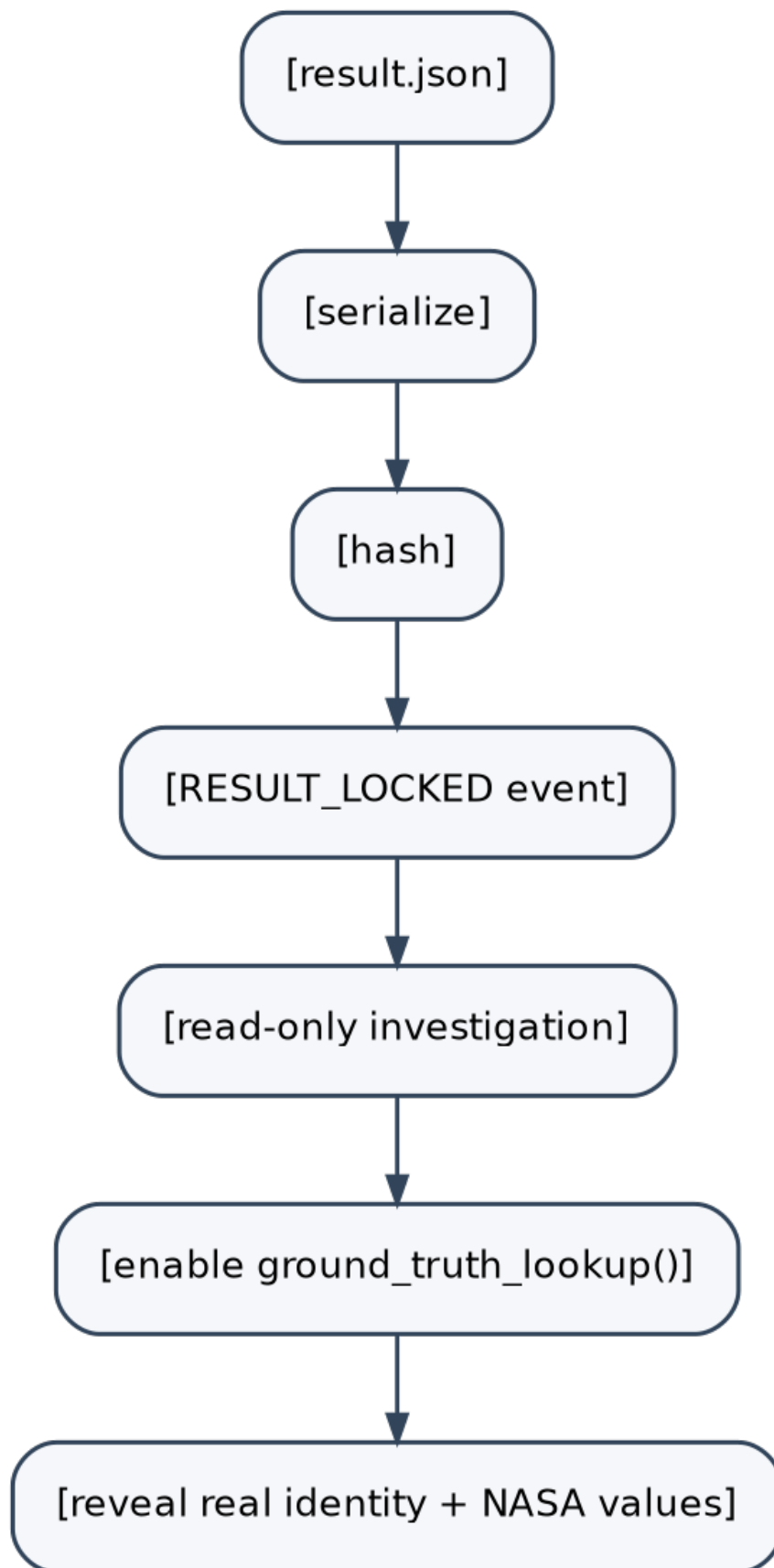
AVAILABLE SERVICES

- cached TESS observation analysis
- permitted contextual summaries
- scientific analysis tools

UNAVAILABLE

- target identity reveal
- NASA known planet parameters
- confirmation status

At finalization:



The trace should prove that `ground_truth_lookup()` was never available before the lock event.
The repository should additionally contain a CI test verifying the restriction.

A package-level dependency rule should prevent the scientific agent layer from directly importing or bypassing the catalog-gating module.

This converts the reveal from storytelling into a legitimate blinded evaluation protocol.

Add a negative control

The demonstration should not only show a known planet.

That leaves a skeptical judge free to say:

"Of course every agent eventually approved it - you picked a planet."

Therefore ExoSwarm should also run the **same architecture, same tool registry and same orchestration rules** on a known false positive, eclipsing binary or similarly non-planetary transit-like signal.

The important result is that the system behaves differently because the evidence differs.

For example:

TARGET B

Transit Hunter:
periodic transit-like signal detected

Skeptic:
odd/even depths inconsistent at significant level

Selected experiment:
test period at 2P

Result:
primary and secondary eclipses separate

Hypothesis update:
eclipsing-binary interpretation strengthened

FINAL DISPOSITION:
planetary interpretation rejected / strongly disfavored

Now we can say:

The Skeptic isn't a ceremonial reviewer. The same system that retained the planetary interpretation in one case rejected or downgraded another transit-like signal because the evidence drove it down a different investigation path.

The two targets must not branch because their hidden names are hard-coded.

They must branch because their evidence states differ.

The main difficulties

False positives

This remains the main scientific difficulty.

A transit candidate is not a planet.

Response: make competing explanations and falsification central rather than hiding the limitation.

Messy astronomical observations

Real TESS observations contain:

- gaps
- outliers
- stellar variability
- spacecraft/systematic effects
- contamination
- differing sectors and cadences

Response: do not promise arbitrary-star robustness.

Use a small target suite whose behavior is understood.

The system should still detect and display data-quality issues rather than silently assuming ideal observations.

For the core hackathon build, avoid multi-sector stitching and broad arbitrary-target support.

Contamination analysis

This is more difficult than a simple nearby-star lookup.

Response: implement one genuine pixel/centroid localization workflow using a cached TESS Target Pixel File.

Do not attempt a full professional validation framework.

Agent reliability

Simply prompting "You are the Skeptic Agent" does not create scientific skepticism.

Response:

- bounded objective
- restricted tool list
- mandatory baseline diagnostics
- structured outputs
- explicit hypothesis state
- Critic review
- eval cases
- code-enforced maximum turns

- different scientific targets producing different trajectories
- structured fallback behavior

Recursive analysis loops

A model could repeatedly request low-value experiments.

Response: every adaptive decision records its expected discriminating value and the runtime tracks completed experiments.

The Critic can veto a redundant proposal.

The Director can terminate when:

- no unused high-value experiment remains
- mandatory tests are complete
- evidence is sufficient for the available disposition categories
- the budget is exhausted
- the agent is repeating itself

Scope creep

A full exoplanet-validation framework is far beyond a weekend.

The prototype proves:

agent-directed transit investigation using a limited, genuine scientific test suite

It does not reproduce every method used by professional exoplanet teams.

Scientific core

The must-have scientific path is:

- cached TESS data
- detrending
- BLS
- phase folding
- period/depth/duration
- odd/even test
- secondary-eclipse test
- P/2-P-2P harmonic test
- one real spatial diagnostic

Agent core

The must-have agent architecture is:

- structured `InvestigationState`
- explicit competing hypotheses
- bounded Skeptic experiment selection

- bounded Critic review
- tool validation
- max turns
- Evidence Ledger
- stopping rules
- opaque target identity
- result lock
- catalog isolation

Demonstration core

The must-have proof is:

- one confirmed-planet case
- one false-positive / eclipsing-binary case
- visibly different agent trajectories
- one visible adaptive experiment decision
- result lock
- NASA reveal
- blind-protocol CI proof
- reproducible repository artifacts

Stretch features

Only after the above works reliably:

- additional models/model routing
- additional agents
- broader target coverage
- advanced probabilistic validation
- extra diagnostics
- multi-sector stitching
- more sophisticated transit fitting

Scientific claims

The project becomes scientifically vulnerable if we say:

"ExoSwarm autonomously discovers confirmed planets."

Do not say this.

Use:

- transit-like signal detection
- candidate investigation
- scientific vetting
- independent blinded recovery

- evidence-based planetary hypothesis
- planetary interpretation survives implemented vetting

When the NASA reveal says:

CONFIRMED PLANET

that status belongs to the NASA catalog, not to ExoSwarm.

Correct language:

ExoSwarm independently recovered the signal and found that the planetary interpretation survived its implemented vetting. Only afterward did the system reveal that NASA lists the object as a confirmed planet.

The novelty claim should remain focused on:

adaptive, adversarial experiment orchestration

not the invention of transit detection, automated vetting or planet confirmation.

The modifications that materially improve ExoSwarm

These should be treated as part of the core specification unless explicitly marked as stretch goals.

1. Make the blind protocol genuinely blind

Agents receive only an opaque target identifier.

The deterministic backend privately maps it to the real target.

Known parameters and real identity remain inaccessible until after result lock.

The blindness claim is verified in CI and the trace.

Impact: HIGH **Implementation risk:** LOW

2. Demonstrate both a confirmed planet and a false-positive case

The two cases must generate visibly different scientific trajectories and final dispositions.

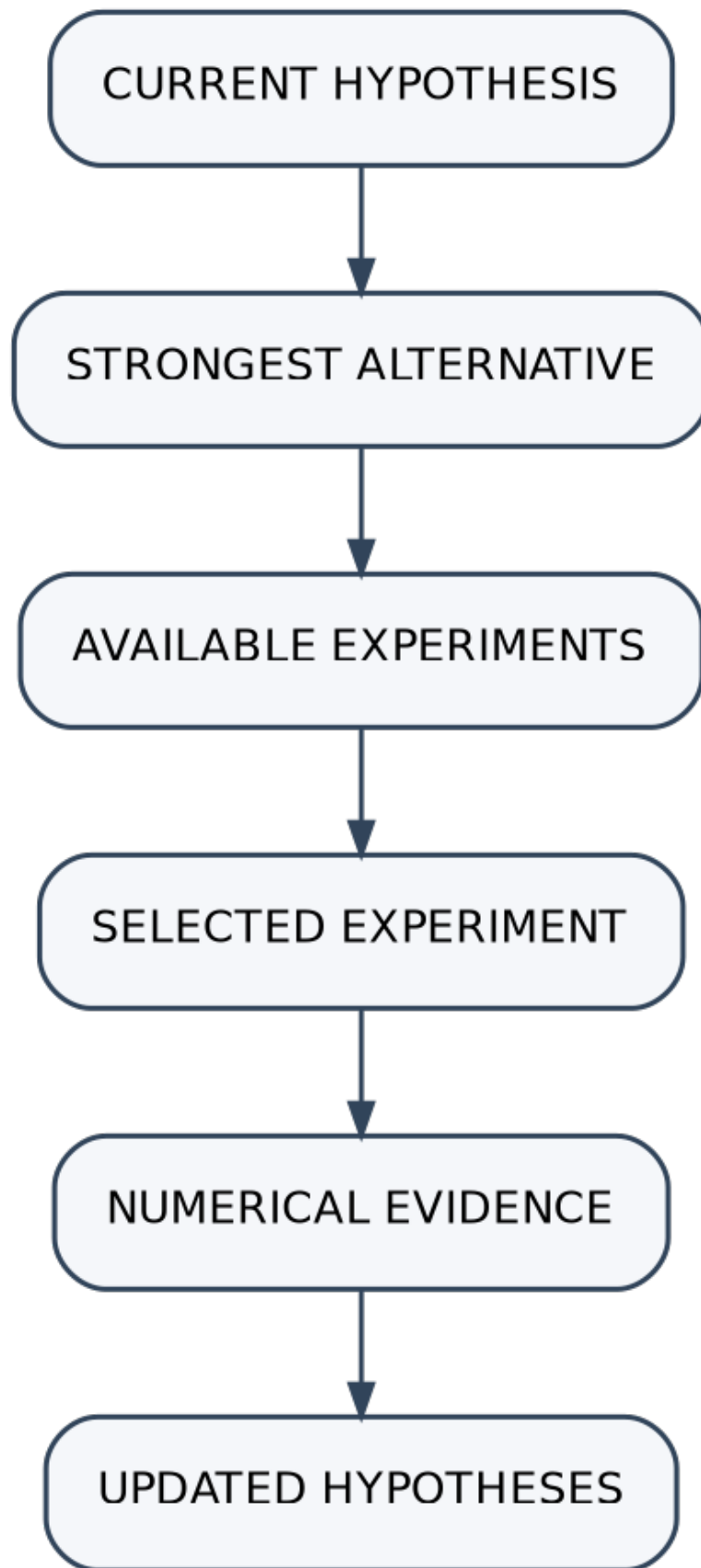
Impact: HIGH **Implementation risk:** LOW-MEDIUM

3. Implement explicit adaptive experiment selection

Mandatory checks remain structural.

Additional tests are chosen from the current evidence.

The UI must make this visible:



Impact: HIGH **Implementation risk:** MEDIUM

4. Add bounded Critic review

Adaptive experiment proposals are checked for redundancy and discriminating value before execution.

A visible VETO or REVISE event in a genuine investigation trace provides especially strong proof that the agent layer is not ornamental.

Impact: HIGH **Implementation risk:** MEDIUM

5. Add an agent-vs-fixed-policy ablation

Compare the adaptive selector against a simple fixed checklist over a small curated eval suite.

Measure actual outcomes, efficiency and consistency rather than merely describing the architecture as agentic.

Impact: HIGH **Implementation risk:** MEDIUM

6. Remove arbitrary AI confidence percentages

Replace them with measured statistics, explicit evidence states, disposition labels and unresolved alternatives.

Impact: HIGH **Implementation risk:** LOW

7. Implement one real pixel/centroid contamination test

This materially strengthens the scientific layer and differentiates the prototype from a toy light-curve demo.

Impact: HIGH **Implementation risk:** MEDIUM

8. Add scientifically honest measurement uncertainty

Period, depth, duration and epoch should include uncertainty or explicit tolerance information where the implemented method allows it.

Impact: MEDIUM-HIGH **Implementation risk:** MEDIUM

9. Cache everything required for the demonstrated investigations

The video and make reproduce path should not depend on live astronomy-data network calls.

Source-data provenance remains attached to every cached artifact.

Impact: HIGH **Implementation risk:** LOW

10. Keep scientific claim language precise

Passing the implemented photometric vetting is not equivalent to confirming a planet.

Every written and spoken claim should remain valid under scrutiny from an astronomer or scientific-computing judge.

Impact: HIGH **Implementation risk:** LOW

How it should look

The original visual direction is excellent and does not need fundamental redesign.

The interface should feel like autonomous scientific mission control, not another dashboard or chat interface.

Main screen

Center

Target star with a candidate orbital visualization.

The orbit and planet become progressively defined as evidence accumulates.

Before reveal, the target is identified only as something like:

UNKNOWN TARGET – TARGET-X17

Scientific panels

Show:

- raw light curve
- cleaned light curve
- BLS periodogram
- phase-folded signal
- odd/even diagnostic
- secondary-eclipse diagnostic
- pixel/centroid view when requested

Agent panel

OBSERVER
complete

SIGNAL
complete

TRANSIT HUNTER
candidate detected

SKEPTIC
investigating contamination

CRITIC
reviewing selected experiment

SCIENTIFIC DIRECTOR
waiting

Do not fill the screen with agent conversations.

Investigation timeline

Example:

```
+ cached TESS observation loaded
+ signal quality accepted

+ periodic signal detected
  P = 6.26812 d +/- ...

+ odd/even test
  consistent

+ secondary-eclipse test
  no significant detection

! nearby source detected

> Skeptic selected:
  centroid localization

> Critic:
  APPROVED - high discriminating value

+ centroid result recorded

o final disposition pending

[LOCKED]
NASA ground truth inaccessible
```

A different target can show:

```
> Skeptic proposed:
  repeat secondary search

> Critic:
  VETO - redundant with existing evidence

> Revised experiment:
  harmonic test at 2P
```

This is strong visual evidence that the agent layer is doing more than narrating a fixed chain.

Evidence board

Replace the original `Confidence: HIGH` panel.

PLANETARY HYPOTHESIS

- + strong periodicity
- + repeated consistent events
- + odd/even depths consistent
- + no significant secondary event
- + centroid consistent with target

REMAINING ALTERNATIVES

- low-level stellar/systematic explanation
not fully excluded

CURRENT DISPOSITION

PLANETARY INTERPRETATION

SURVIVES IMPLEMENTED VETTING

If deterministic evidence-weight updates are implemented, hypothesis bars can visibly move as evidence arrives, but they should not be labeled as calibrated probabilities unless they genuinely are.

Agent decision panel

This is extremely important.

Whenever an adaptive branch occurs, display:

SKEPTIC

Current hypothesis:
PLANETARY TRANSIT

Strongest unresolved alternative:
BACKGROUND CONTAMINANT

Available experiments:

- centroid localization
- alternate aperture test
- stop

Selected experiment:
CENTROID LOCALIZATION

Why:
A nearby source is present inside the aperture.
Spatial evidence is now more discriminating than another photometric test.

Expected outcome:
This experiment should distinguish whether the dimming is spatially associated with the target.

Then show the Critic:

CRITIC

Verdict:
APPROVE

Reason:
The experiment addresses the strongest remaining alternative and is not redundant with existing evidence.

That makes the advanced-inference architecture understandable immediately.

The ideal 3-minute video

The original three-minute structure is strong, but the presentation should be designed specifically for an **online recorded submission**, not a live stage demonstration.

Everything important should be legible without narration.

Use burned-in captions and large on-screen states because judges may watch muted, skip around, or use increased playback speed.

There is no meaningful distinction between "live" and "pre-recorded" inside the submitted video, so do not spend time pretending otherwise.

Show the strongest reproducible evidence.

0:00-0:15 - Hook

Mission control is already visible.

Large text:

UNKNOWN TESS TARGET
TARGET-X17

TARGET IDENTITY
[LOCKED]

NASA PLANETARY PARAMETERS
[LOCKED]

Caption/voiceover:

This is real NASA TESS data. The agents have not been given the target identity or known planetary parameters. ExoSwarm's job is not just to find a signal - it has to search for evidence that could prove its own planetary hypothesis wrong.

Immediately begin the investigation.

Do not start with explanatory slides.

0:15-0:55 - Deterministic search

Show:

- cached NASA/TESS observation loading

- data-quality result
- preprocessing selection
- cleaned light curve
- BLS periodogram
- phase-folded signal
- measured period/depth/duration/SNR with uncertainty or tolerance

Large state:

TRANSIT-LIKE SIGNAL FOUND

Make it visually obvious that deterministic tools produced the measurements.

0:55-1:25 - The differentiator

Show the current hypothesis state:

CURRENT HYPOTHESIS
planetary transit

STRONGEST COMPETING EXPLANATION
background contaminant

AVAILABLE EXPERIMENTS
centroid localization
alternate aperture test
stop

SKEPTIC SELECTED
centroid localization

Then show the Critic.

Where a curated genuine trace contains one, a VETO or REVISE event is especially useful:

CRITIC
VETO

Reason:
proposed test duplicates existing evidence

Suggested revision:
centroid localization

Run the deterministic spatial diagnostic.

Show the image or centroid result.

Show the Evidence Ledger updating.

Caption:

The LLM isn't calculating these values. It is deciding what evidence the investigation should seek next. Every scientific measurement comes from deterministic astronomy code.

1:25-1:55 - Negative control

Do not reduce the false-positive case to a barely visible aside.

Give it enough time to prove that the system behaves differently.

Show a second opaque target:

TARGET-X42

periodic signal detected
odd/even inconsistency found

Skeptic:
test period at 2P

Critic:
APPROVE

Result:
primary and secondary events separate

UPDATED HYPOTHESIS:
eclipsing binary strengthened

FINAL DISPOSITION:
PLANETARY INTERPRETATION REJECTED / STRONGLY DISFAVORED

Caption:

The same system does not always end at "planet." Different evidence caused a different experiment sequence and a different conclusion.

1:55-2:10 - Prove the blindness in code

Briefly show the repository/terminal.

For example:

```
pytest tests/test_blind_protocol.py  
PASS
```

```
architecture/import boundary  
PASS
```

Large caption:

Blindness is a software invariant, not a promise.

This is particularly valuable in an offline hackathon because judges can inspect the same test after watching the video.

2:10-2:35 - Lock and reveal

Return to the primary target.

Show:

EXOSWARM RESULT

Period:
6.26812 d +/- ...

Epoch:
... +/- ...

Duration:
... +/- ...

Depth:
... +/- ...

Disposition:
PLANETARY INTERPRETATION
SURVIVES IMPLEMENTED VETTING

TARGET IDENTITY:
[STILL LOCKED]

NASA PARAMETERS:
[STILL LOCKED]

Then:

SERIALIZING RESULT...
HASH CREATED
RESULT LOCKED

The interface becomes read-only.

Only now:

UNLOCK TARGET + NASA CATALOG

Show:

	EXOSWARM	NASA
Period	6.26812 d	6.26794 d
Epoch	...	...
Depth	...	...

Period error:
0.0029%

NASA DISPOSITION:
CONFIRMED PLANET

Make the distinction explicit:

ExoSwarm's implemented vetting did not confirm the planet. NASA's catalog independently lists the revealed object as confirmed.

2:35-2:50 - Show that the agent claim is evaluated

Briefly show the eval/ablation report.

Useful visible rows:

FIXED CHECKLIST vs AGENT-DIRECTED

Correct disposition
Mean experiments
Unnecessary experiments
Invalid tool requests
Realized information value / experiment
pass^3 consistency

Do not fabricate favorable numbers.

Use the actual measured results.

Caption:

We evaluate whether the agent changes the scientific trajectory, not whether its explanation sounds intelligent.

2:50-3:00 - Close on a forward prediction

The NASA comparison demonstrates blinded recovery.

The strongest final visual is then a genuine forward prediction derived from the locked ephemeris:

NEXT TRANSIT PREDICTION
<time> +/- <uncertainty>

The exact time must come from the deterministic ephemeris computation.

Closing line:

ExoSwarm didn't know the answer. It measured the signal, chose experiments to challenge its own hypothesis, locked the result, and only then saw the catalog.

Then:

The AI doesn't replace the scientific tools. It decides which scientific question to ask next.

End there.

The points to emphasize if you want to win

1. This is real science, not an LLM simulation of science

Show:

- real cached TESS observations
- real numerical algorithms
- real diagnostic calculations

- real external ground truth
- reproducible source provenance

The LLM never fabricates the period.

2. The agents are necessary because their decisions change the computation

Do not focus on:

"We have five agents."

Focus on:

The evidence determines which experiment the system runs next.

If target A causes:

`centroid_localization()`

while target B causes:

`test_period_harmonic()`

because their evidence differs, that is much stronger proof of agentic behavior than several chat bubbles.

3. The AI actively attempts to falsify itself

This remains the strongest conceptual differentiator.

Do not say:

"Five AI agents collaborate."

Say:

The system actively searches for evidence that would make its current explanation fail.

That is both more interesting and more scientifically meaningful.

4. Give judges quantitative external proof

Do not finish with:

"Our AI says confidence = 94%."

Show:

We predicted X before accessing the catalog. NASA reports Y.

External ground truth beats self-reported AI confidence.

5. Make the blindness mechanically verifiable

The offline format allows the strongest version of this argument.

A judge can inspect:

- the blind-protocol test
- the package boundary
- the result hash
- the trace
- the reveal artifact

The system does not ask the judge to trust the video.

6. Prove the Skeptic changes outcomes

The negative-control target is essential.

Show that the same architecture can:

- retain the planetary hypothesis on one target
- reject or downgrade it on another

That turns the Skeptic from a narrative device into a real component.

7. Prove the adaptive experiment is reviewed

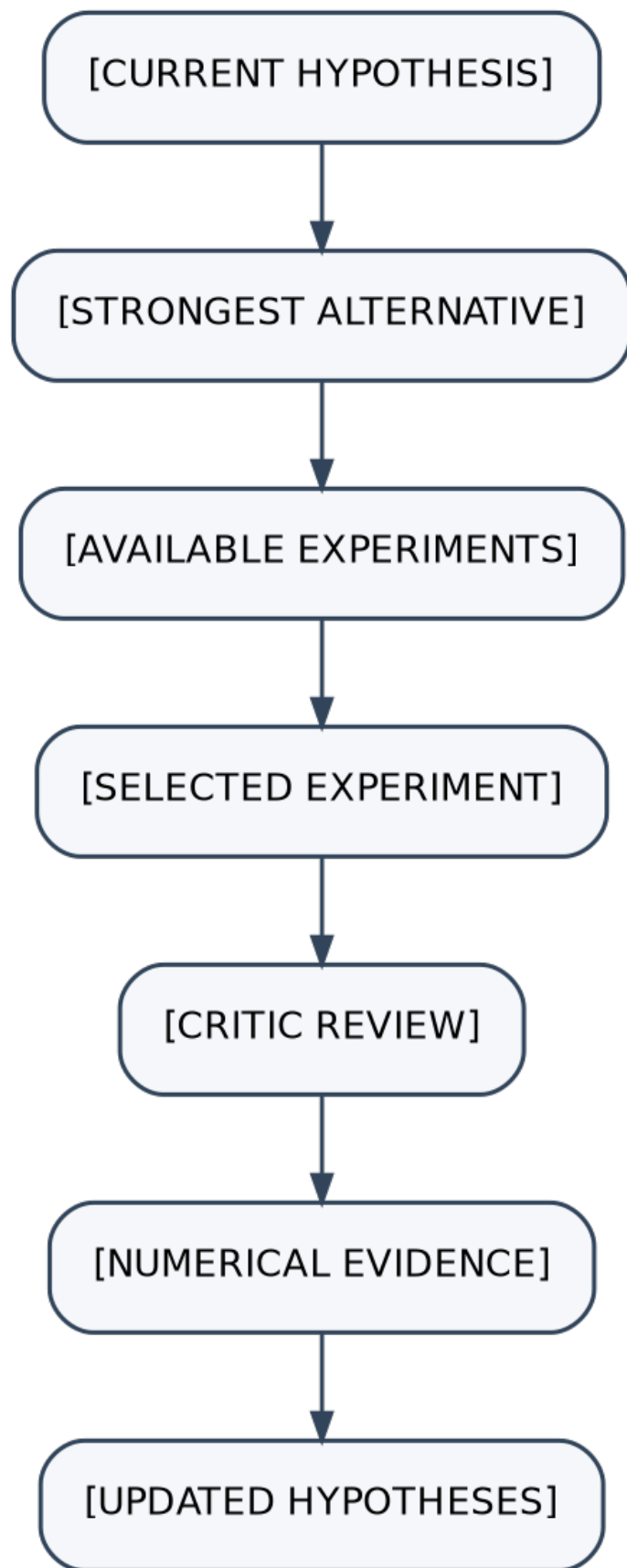
The Critic gives a second layer of visible evidence that experiment selection is not ornamental.

An actual `APPROVE`, `REVISE` or `VETO` event demonstrates that the proposal is evaluated against the scientific ledger before computation.

8. Make the inference loop visually obvious

A judge should be able to understand the system without seeing the code.

The interface itself should expose:



That is the advanced inference pipeline.

9. Evaluate the agent rather than merely describing it

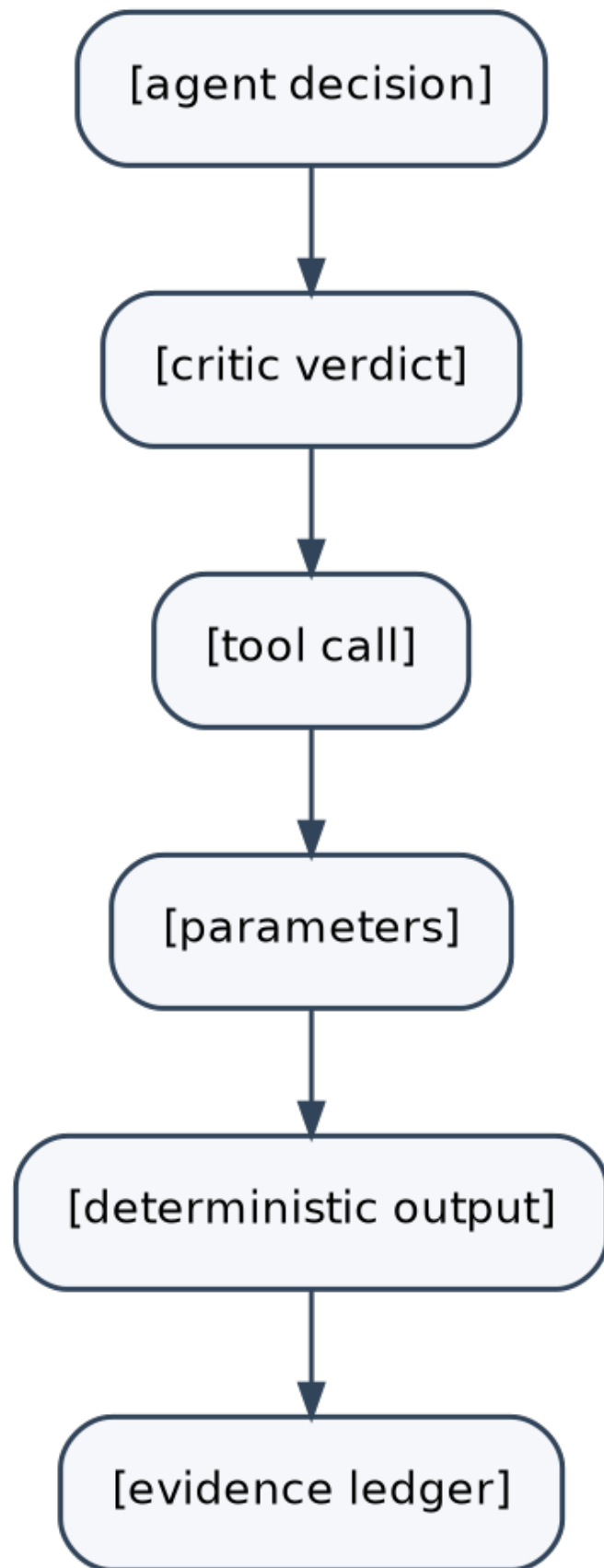
The ablation and repeated eval runs directly answer:

"Why isn't this just an if/else pipeline?"

Show actual measured trajectory differences, unnecessary-tool counts, final dispositions, information value and consistency.

10. Make the architecture auditable

Every important result should trace back to:



Every catalog reveal should occur after `RESULT_LOCKED`.

Technical judges will appreciate this disproportionately.

11. Do not oversell the wrong thing

The impressive claim is not:

"We invented exoplanet detection."

We did not.

It is not:

"We are the first agentic exoplanet system."

That is not a defensible novelty claim given existing work.

The defensible claim is:

We turned established scientific computation into a visible, adversarial investigation loop where AI agents choose which experiment to run next, while every scientific measurement remains deterministic, auditable and independently checkable.

Priority order for the hackathon

P0 - Cannot ship without

- reliable cached TESS -> BLS -> folded transit pipeline
- one planet case
- one false-positive / eclipsing-binary case
- different evidence-driven agent paths
- anonymous target identity
- technically enforced result lock
- no fake confidence
- precise scientific claim language
- Evidence Ledger
- bounded adaptive experiment selection
- cached demonstration data
- reproducible traces
- blind-protocol test

P1 - Major competitive advantage

- real pixel/centroid contamination test
- visible hypothesis updates
- Critic review
- agent-vs-fixed-policy mini evaluation
- repeated-run consistency metric
- value-of-information logging
- numeric-provenance guardrail

P2 - Only after everything works

- improved transit fitting and uncertainties beyond the basic required level
 - multi-model routing
 - broader target coverage
 - additional diagnostics
 - additional agents
 - more sophisticated probabilistic validation
 - multi-sector stitching
-

Target demonstration flow after all fixes

1. UNKNOWN TARGET

Real cached TESS observations load.

Agents receive only an opaque target ID.

NASA ground truth and real identity are visibly locked.

2. DETERMINISTIC SEARCH

Detrending, BLS and phase folding produce the candidate measurements.

3. HYPOTHESIS STATE

Planet, eclipsing binary, contamination, variability/systematics and alias alternatives are visible.

4. SKEPTIC DECISION

The agent selects the most discriminating available experiment and explains why.

5. CRITIC REVIEW

A bounded evaluator checks whether that experiment is genuinely informative rather than redundant.

6. REAL COMPUTATION

Python executes the test.

The Evidence Ledger records exact inputs, outputs, uncertainties and interpretation codes.

7. NEGATIVE CONTROL

A second target follows a different evidence-driven branch and is downgraded or rejected.

8. BLINDNESS PROOF

The repository shows that target identity and ground-truth lookup are mechanically unavailable before lock.

9. RESULT LOCK

ExoSwarm serializes and hashes its measurements before any ground-truth service is enabled.

10. NASA REVEAL

Only after lock are the real target identity and NASA values exposed.

ExoSwarm's own implemented vetting remains clearly separated from NASA confirmation status.

11. EVALUATION PROOF

The agent-directed policy is compared against a simple fixed-policy baseline across a curated suite.

12. FORWARD PREDICTION

The locked ephemeris produces a next-transit prediction with scientifically honest uncertainty.

Final defensible claim

ExoSwarm is an AI-orchestrated TESS investigation system that uses deterministic astronomy tools to test and falsify competing explanations for transit-like signals, adaptively chooses which scientific evidence to seek next, and locks its measurements before comparing them with NASA ground truth.

The project is not attempting to outperform professional exoplanet pipelines at detecting planets.

It demonstrates something different:

an AI-controlled scientific investigation layer capable of deciding what evidence to seek next, challenging its own working hypothesis, and producing a reproducible scientific trail whose numerical conclusions can be independently checked.

It preserves everything strong about the original idea while directly addressing its largest weaknesses:

- agents must affect the scientific trajectory
- the Skeptic must be capable of changing the outcome
- adaptive choices must be reviewable rather than ornamental
- numerical confidence cannot be invented
- final measurements should communicate uncertainty
- contamination analysis must be scientifically meaningful
- ground truth and recognizable target identity must be genuinely inaccessible before lock

- context must remain compact and evidence-driven
- the agent loop must be bounded and observable
- agent failure behavior must be explicit
- the project must demonstrate both successful blinded recovery and meaningful rejection of a false-positive case
- the agentic claim must be supported by evals and an ablation rather than presentation alone
- the online video must remain understandable without a live presenter
- the repository must allow a skeptical judge to verify the central claims without trusting the narration

The resulting ExoSwarm is therefore not an "AI planet detector."

It is **AI mission control for a falsifiable, auditable scientific investigation.**